

A PROJECT TITLE

ON

Player Performance Prediction Using XGBoost and Gradient Boosting
project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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DEPARTMENT OF JCSIT

Name of Title: Player Performance Prediction Using XGBoost and Gradient Boosting

1. Abstract:

Player performance prediction has become one of the most significant applications of machine learning in modern sports analytics. Accurate prediction of an athlete's future performance enables coaches, selectors, analysts, and team management to make informed decisions regarding team selection, training strategies, player development, and match planning. Traditional methods of evaluating player performance primarily depend on expert opinions and historical statistics, which may not effectively capture complex relationships among multiple performance factors. To overcome these limitations, this project proposes a machine learning-based player performance prediction system using **Gradient Boosting** and **Extreme Gradient Boosting (XGBoost)** algorithms.

The proposed system utilizes historical player performance data containing various statistical features such as matches played, innings, runs scored, batting average, strike rate, centuries, wickets taken, bowling average, economy rate, catches, and other relevant performance indicators. The collected dataset undergoes several preprocessing steps, including data cleaning, handling missing values, feature selection, normalization, and encoding of categorical variables to improve data quality and model performance. After preprocessing, the dataset is divided into training and testing sets, and both Gradient Boosting and XGBoost models are trained to predict player performance accurately.

The performance of the models is evaluated using standard machine learning metrics such as Accuracy, Precision, Recall, F1-Score, Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE), depending on the prediction task. Experimental results indicate that XGBoost generally outperforms the traditional Gradient Boosting algorithm by providing higher prediction accuracy, faster execution, better handling of missing values, and reduced overfitting through regularization techniques. The developed system demonstrates the

effectiveness of ensemble learning methods in sports analytics by delivering reliable and data-driven predictions. The proposed model can assist sports organizations, coaches, and performance analysts in identifying promising players, optimizing team composition, monitoring player progress, and making strategic decisions. Furthermore, the framework can be extended to various sports such as cricket, football, basketball, and hockey, making it a versatile and scalable solution for intelligent player performance prediction.

2. INTRODUCTION:

In recent years, sports organizations have increasingly adopted data analytics to improve team performance and decision-making. Traditional player evaluation methods often rely on expert judgment and historical observations, which may not always provide accurate predictions. Machine learning techniques offer a more reliable approach by analyzing large volumes of player statistics and identifying hidden patterns within the data.

This project focuses on predicting player performance using **Gradient Boosting** and **XGBoost**, two advanced ensemble learning algorithms that combine multiple decision trees to produce highly accurate predictions. Gradient Boosting builds models sequentially by minimizing prediction errors, while XGBoost enhances this process with regularization, parallel processing, and optimized computation.

The system involves data collection, preprocessing, feature engineering, model training, prediction, and performance evaluation. By comparing the results of both algorithms, the project identifies the most suitable model for predicting player performance. The proposed system can be applied in cricket, football, basketball, and other sports where player statistics are available. It assists coaches and management in selecting players, planning strategies, and improving overall team performance through data-driven insights.

3. SOFTWARE REQUIREMENTS:

Category	Requirement
Operating System	Windows 10/11, Ubuntu Linux, or macOS

Category**Requirement**

Programming
Language

Python 3.10 or later

IDE

Jupyter Notebook, Visual Studio Code, PyCharm, or Google Colab

Machine Learning
Libraries

Scikit-learn, XGBoost

Data Processing

Pandas, NumPy

Data Visualization

Matplotlib, Seaborn

Model Evaluation

Scikit-learn Metrics

Dataset Format

CSV (.csv)

Version Control
(Optional)

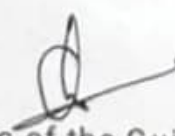
Git, GitHub

Browser

Google Chrome, Microsoft Edge, Firefox

Hardware

Intel Core i3 or above, 4 GB RAM (8 GB recommended), 10 GB free disk space


Signature of the Guide

Course Instructor